SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –Pseudocode Design**	**Assessment:**	**Assessment Tool 1– Pseudocode Design**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
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Section / Batch:	2023/AI&DS	**Date:**	24.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Design pseudocode to classify a new student as Pass or Fail using kNN based on attendance percentage, internal marks, and assignment scores.

1. Problem Statement

Design a **k-Nearest Neighbors (kNN) classifier** to classify a new student as **Pass** or **Fail** based on:

- Attendance Percentage
- Internal Marks
- Assignment Score

The classifier compares the new student's data with previously recorded students and assigns the class based on the majority class among the **K nearest students**.

2.Input

Training Dataset:

Student	Attendance %	Internal Marks	Assignment Score	Result
S1	85	78	80	Pass
S2	60	45	50	Fail
S3	90	85	88	Pass
S4	55	40	45	Fail
S5	75	70	72	Pass

New Student:

- Attendance = 80%
- Internal Marks = 75
- Assignment Score = 78
- K = 3

Output: Pass or Fail

kNN is a **supervised machine-learning classification algorithm**.
It classifies a new data point by:

1. Calculating the distance between the new student and every training student.

2. Sorting the distances from smallest to largest.
3. Selecting the **K nearest neighbors**.
4. Counting the number of Pass and Fail classes.
5. Assigning the class with the majority vote.

The commonly used **Euclidean distance** is:
where:

$$d = \sqrt{(A_1 - A_2)^2 + (M_1 - M_2)^2 + (S_1 - S_2)^2}$$

- = Attendance
- = Internal Marks
- = Assignment Score

3. Pseudocode

ALGORITHM: kNN Student Pass/Fail Classification
START
1. Read the training dataset.
2. For each student, store:
    Attendance
    Internal Marks
    Assignment Score
    Result (Pass/Fail)
3. Read the new student's:
    Attendance
    Internal Marks
    Assignment Score
4. Choose the value of K.
    K = 3
5. FOR each student in the training dataset DO
    Calculate Euclidean distance:
    Distance =
    sqrt(
      (New_Attendance - Student_Attendance)^2+
       (New_Internal - Student_Internal)^2 +(New_Assignment - Student_Assignment)^2
)
    Store the calculated distance along with the student's Result.

END FOR
6. Sort all students according to their distance
in ascending order.
7. Select the first K students.
8. Count:
    Pass_Count = number of Pass students
    Fail_Count = number of Fail students
9. IF Pass_Count > Fail_Count THEN
    Prediction = "PASS"
ELSE
    Prediction = "FAIL"
END IF
10. Display Prediction.

END

4. Example Calculation

For the new student:

(80, 75, 78)

For S1 **(85, 78, 80)**:

The same calculation is performed for all training students.

After sorting the distances, the **3 nearest students** are selected. If 2 are Pass and 1 is Fail:

Therefore:

Prediction = PASS

5. Requirements

- Training dataset containing previous student records
- Attendance percentage
- Internal marks
- Assignment scores
- Result labels: Pass/Fail
- Value of K
- Euclidean distance calculation
- Sorting mechanism

- Majority voting

Flow of the Algorithm